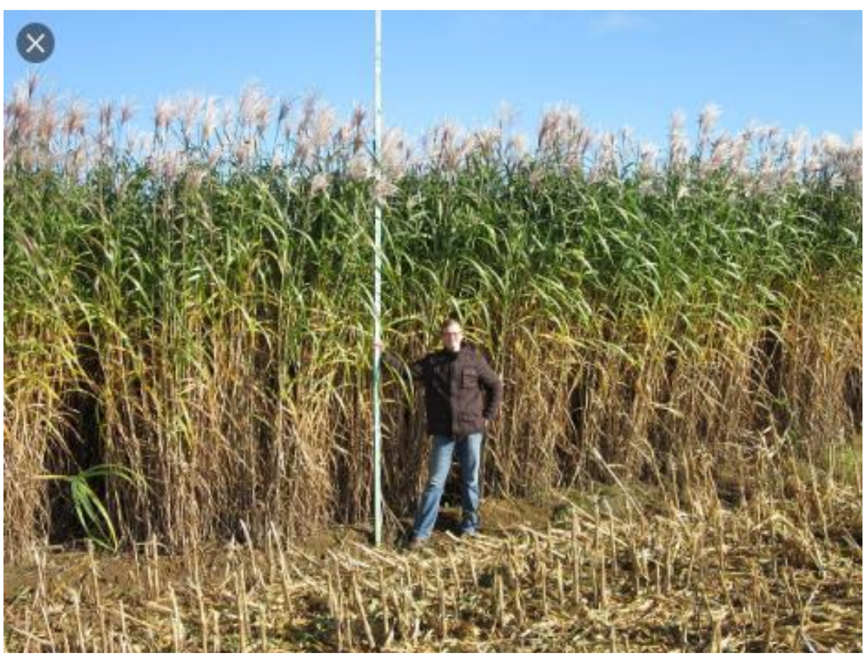
NUMERICALS W.R.T. BIOMASS CHARACTERISTICS

➢ *Miscanthus × giganteus*, the giant miscanthus (Figure-1)

- **It is a sterile hybrid of *Miscanthus sinensis* and *Miscanthu sacchariflorus*.**
- **It is a perennial grass with bamboo-like stems that can grow to heights of more than 4 metres (13 ft) in one season (from the third season onwards).**
- **It is also called elephant grass.**
- **It has ability to grow on marginal land.**
- **Its water efficiency, non-invasiveness, low fertilizer needs, significant carbon sequestration and high yield have sparked a lot of interest among researchers.**
- **It is known as energy crop.**
- **Commercially grown in the European Union, mainly due to its cold weather tolerance.**



- **Grown on large scale by Anglian Straw Company, the world's largest straw fired power plant in England for use alongside wheat straw .**

FIGURE-2: STEEP MARGINAL LAND FOR CROP

FIGURE-3: TRANSPORT OF BULKY, WATER ABSORBING MISCANTHUS BALES IN ENGLAND.

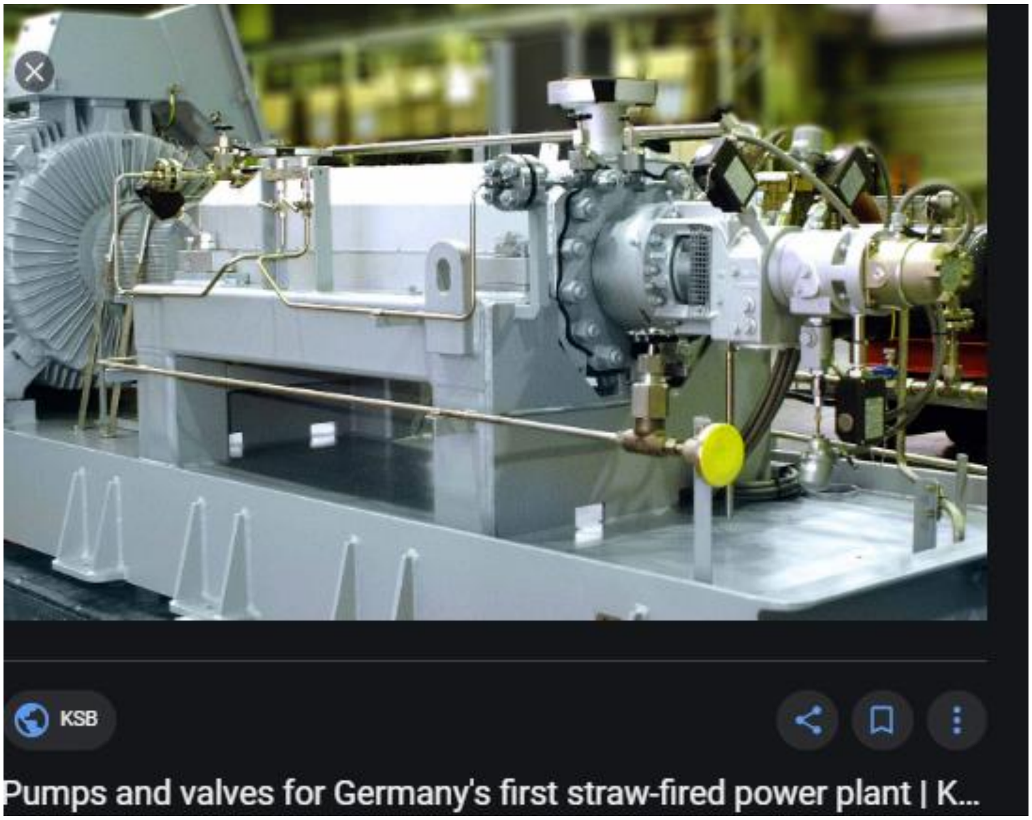


FIGURE-4

FIGURE-5: GOOLE STRAW PELLET PLANT (2009).

NUMERICAL PROBLEMS

➢ DETERMINATION OF HIGHER HEATING VALUES

- **Higher heating Value is one of the most imortant properties of fuels which explain the higher energy content and determine the efficient use of biomass and fossil fuels**
- **HHV comes to design calculations or numerical simulation of thermal conversion systems of biofuels.**
- **There are numerous equations proposed in the literature for calculating the HHV of fuels from the basic analysis data.**
- **The proximate and ultimate analysis of fuels are necessary for their efficient and clean utilization while the HHV determine the quantitative energy content of fuels.**
- ***No.1: Miscanthus* × *giganteus* ultimate analysis on mass basis (as received) is given as:**

	As received	Dry basis	Dry ash free basis
Carbon	43.10		
Hydrogen	5.27		
Sulfur	0.12		
Nitrogen	0.34		
Oxygen	40.43		
Ash	2.23		
Moisture	8.51		
Total	100		

Calculate on dry and dry ash free basis

- **No.2: Calculate chemical formula subscript for *Miscanthus* × *giganteus* with the help of following information:**

Elements	Wt.%	Atomic wt.	No. of Atoms
Carbon	47.31	12	
Hydrogen	5.78	1	
Sulfur	0.14	32	
Nitrogen	0.37	14	
Oxygen	44.09	16	
Chlorine	0.04	35	

- **No.3: With the help of the following equation and Table given in Q. No.1, calculate HHV on wet basis, dry basis:**

$$HHV_{\text{wet basis}} = 0.35C + 1.18H + 0.1S - 0.02N - 0.01O - 0.02A$$

- **No.4: Calculate the lower heating value of the feedstock on dry basis using the following co-relation:** $LHV = HHV - \left(\frac{m_{H_2O}}{m_{\text{feed}}}\right)h_{fg} - (8.936H)h_{fg}$**, where** $\frac{m_{H_2O}}{m_{\text{feed}}}$ **is the moisture content of the feedstock from proximate analysis and** h_{fg} **is standard latent heat of vaporization of water at 25 ^{0}C, which is 2.447 MJ/kg.**

- **No. 5: The lignin content and higher heating values (HHVs) of biomass samples are given in the following Table:**

No.5 The lignin contents and higher heating values (HHV) for some of the biomass samples are given in the following Table:

Biomass Samples	Lignin (L) Measured (wt.%)^{db}	HHV (wt.%)^{db} (MJ/kg)	HHV (wt.%)^{daf} (MJ/kg)	HHV (Calculated) (MJ/kg)	Difference
Corn Stover	14.4	17.8	18.5	17.7	-0.8
Corncob	15.0	17.0	17.2	17.8	+0.6
Sunflower Shell	17.0	18.0	18.8	18.0	-0.8
Beech Wood	21.9	19.2	19.5	18.4	-1.1
Ailanthus Wood	26.2	19.0	19.4	18.9	-0.5
Hazelnut Shell	42.5	20.2	20.5	20.1	-0.4
Wood Bark	43.8	20.5	20.8	20.1	-0.7
Olive Husk	48.4	20.9	21.6	21.0	-0.6
Walnut Shell	52.3	21.6	22.2	21.4	-0.8

No.6 With the help of the above data, develop a mathematical model which correlates higher heating values and the lignin contents.

After finding the mathematical correlation between HHV and L in question (1), determine the square of correlation coefficient (R^2) and also calculate percentage of average error, and what is the root mean square error (RMSE)?

Note: The correlation: $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\text{Observed value} - \text{Predicted value})^2}$ may be used.

Measured Value of Lignin (wt%)^{db}	Calculated HHV
14.4	17.7
15	17.8
17	18
21.9	18.4
26.2	18.9